MINISTRY OF EDUCATION, SINGAPORE
in collaboration with
CAMBRIDGE ASSESSMENT INTERNATIONAL EDUCATION
General Certificate of Education Advanced Level
Higher 2

COMPUTING

9569/01

Paper 1 Written

October/November 2025

3 hours

READ THESE INSTRUCTIONS FIRST

An answer booklet will be provided with this question paper. You should follow the instructions on the front cover of the answer booklet. If you need additional answer paper ask the invigilator for a continuation booklet.

Answer **all** questions.
Approved calculators are allowed.

The number of marks is given in brackets [ ] at the end of each question or part question.
The total number of marks for this paper is 100.

1 A company sells components.

- Customers order a number of components.
- The company supplies a maximum of 50 components per order.
- The company arranges delivery if the number of components ordered is no more than its current stock level and the customer has sufficient funds to cover the order cost. If the customer has sufficient funds, but the company does not have the number of components required in stock, the order is put on hold. Otherwise, the order is rejected.

(a) Copy and complete the decision table. Include all possible combinations and do **not** remove redundancies at this point.

		Rules							
Conditions	components ordered <= 50								
	customer has sufficient funds								
	amount in stock < components ordered								
Actions	deliver components								
	hold order								
	reject order								

[4]

(b) Draw a simplified decision table by removing redundancies from your answer in part **(a)**. [2]

(c) Write pseudo-code to input the number of components the customer wants to order and output an appropriate response from:

- 'Order accepted'
- 'Max order size exceeded'
- 'Insufficient stock'
- 'Insufficient funds'.

Use the identifier names in the table in your pseudo-code:

Identifier name	**Description**
`InStock`	number of components currently in stock
`NumOrdered`	number of components customer wants to order
`TotalOrderCost`	the order cost
`CustBalance`	the customer's funds

[5]

(d) Copy and complete the table to give suitable numerical values for `NumOrdered` for the test cases given:

Test case	**Value**
abnormal	
extreme	
normal	

[3]

2 A function `Puzzle` uses iteration.

```
01 FUNCTION Puzzle(N : INTEGER) RETURNS STRING
02    S = ""
03    WHILE N != 0
04       R = N MOD 2
05       N = N DIV 2
06       S = ToString(R) + S
07    ENDWHILE
08    RETURN S
09 ENDFUNCTION
```

`ToString` is a predefined subroutine that converts an integer to a string.

(a) Copy and complete the trace table for the function call `Puzzle(109)`.

You may add as many additional rows to the table as you need.

N	S	R

[3]

(b) Give **one** reason why `R` needs to be converted to a string on line `06`. [1]

(c) Identify the purpose of the function `Puzzle`. [1]

(d) Rewrite the function `Puzzle` as a recursive function. [3]

(e) Identify **two** advantages of using the iterative version of the function `Puzzle` instead of the recursive version. [2]

(f) Both the ASCII and the Unicode value for the character 'n' is 110.

(i) Calculate the hexadecimal value for the denary value 110. [1]

(ii) Identify **two** benefits of using Unicode for encoding characters. [2]

[Turn over

3 An outdoor clothing shop takes orders from customers.

It uses a database to store customers' orders for the products it sells. It also stores its suppliers' details. Suppliers provide the products that the company sells to its customers. No two suppliers provide the same product.

Each customer has:
- a customer number
- a name
- a delivery address
- an email address.

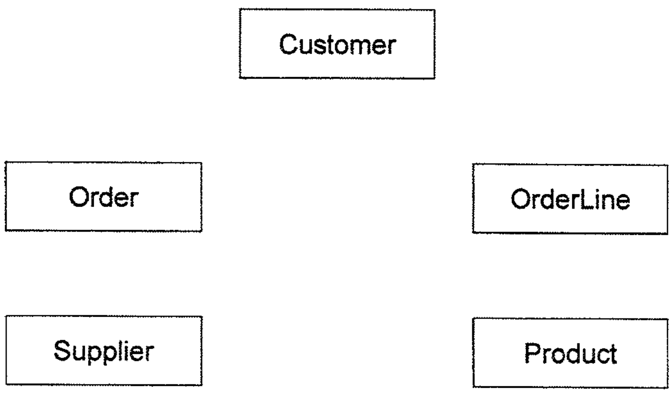
Each supplier has:
- a supplier number
- a name
- an email address
- a telephone number.

Each product has:
- an identification code
- a description
- a retail price
- a current stock level
- a minimum stock level.

Each order has:
- an order number
- a transaction date
- a set of order lines, where each order line specifies the quantity of each product that the customer orders.
  For example, there are two order lines if a customer orders three pairs of gloves and two hats.

(a) The database will have **five** tables.

Copy and complete this entity-relationship (ER) diagram by showing **four** one-to-many relationships. When copying the diagram, do **not** rearrange the layout of the five given entities. If needed, your answer may include lines that cross over each other.



[4]

(b) A table description can be expressed as:
`TableName (Attribute 1, Attribute 2, Attribute 3, …)`

The primary key is indicated by underlining one or more attributes. Foreign keys are indicated by using a dashed underline.

Write table definitions for each of the tables listed.

Do **not** use any compound primary keys.

(i) Customer [1]

(ii) Order [2]

(iii) OrderLine [2]

(iv) Supplier [1]

(v) Product [2]

(c) The company wants a report to display the products that need to be ordered from its suppliers.

Products need to be reordered when the current stock level drops below the minimum stock level.

A report is produced that displays product description, price, supplier name and supplier email address for the items that need reordering.

Write an SQL query to output the required data. [4]

(d) Give **one** reason why a supplier's telephone number should **not** be stored as an integer. [1]

(e) Identify **two** suitable validation checks that could be performed when a customer email address is input. [2]

(f) The company is investigating using a NoSQL Database Management System (DBMS) rather than a relational database.

State **two** reasons why the company may **not** want to use a NoSQL DBMS. [2]

4 A stack is to be implemented as a class.

The stack will be implemented as a static array of 10 string elements.

Part of a class diagram is given:

Stack
`- MaxSize : Integer`
`+ IsEmpty() : Boolean`

Legend

+	Public
–	Private

(a) Copy and complete the class diagram to add the attributes and methods necessary for the stack class, making it clear whether they are public or private. [5]

(b) Describe **two** benefits of using Object-Oriented Programming (OOP) to implement the stack as a class. [4]

(c) Explain how the stack class could be used to reverse a variable length sequence of inputs such as X Y Z to output Z Y X. [3]

(d) A stack could be implemented by using a dynamic linked list.

Identify **two** benefits of using a dynamic structure. [2]

5 A bank currently provides services to its customers via a web application.

(a) Customers type in a Uniform Resource Locator (URL) into a web browser to access the web application.

https://www.newbank.com/onlinebanking/sign-in.html

Copy and complete the table to describe each part of the URL.

Part of URL	Description
https	
www	hostname
newbank.com	
onlinebanking/sign-in.html	

[3]

(b) Explain how the IP address of the bank's server is found when the URL is entered into a web browser. [2]

(c) The bank uses an intrusion detection system (IDS) as part of its security measures.

State **three** features of an IDS. [3]

(d) The bank wants to develop a new native application for its customers to access their accounts from mobile phones.

(i) Explain what is meant by a native application. [2]

(ii) Identify **two** benefits of developing a native application. [2]

(e) The software team that develops the application also writes technical documentation.

They use version control with naming conventions for the technical documentation.

(i) Give **two** benefits of using version control. [2]

(ii) Explain the need for naming conventions. [2]

6 A programmer has implemented two sorting algorithms: bubble sort and merge sort.

The contents of an array are shown:

9	6	1	3	7

(a) Use annotated diagrams to show how a bubble sort algorithm will sort the data in the array into ascending order. [3]

(b) Use annotated diagrams to show how a merge sort algorithm will sort the data in the array into ascending order. [3]

(c) Using Big-O notation, state the worst-case time complexity of each sorting algorithm in the table.

Copy and complete the table:

Sorting algorithm	Worst-case time complexity
bubble sort	
merge sort	

[2]

(d) Identify **two** reasons why a bubble sort might be used instead of a merge sort. [2]

(e) A binary search algorithm can be used to search for a value in an array.

(i) State **one** precondition required for a binary search to work. [1]

(ii) Describe the steps in a binary search. [5]

7 Autonomous taxis are unmanned vehicles that operate using Artificial Intelligence (AI). These taxis make decisions independently. For example, they navigate the quickest and safest route to pick up and drop off passengers at their destination.

Discuss the ethical issues that may be raised by the introduction of autonomous taxis. [6]